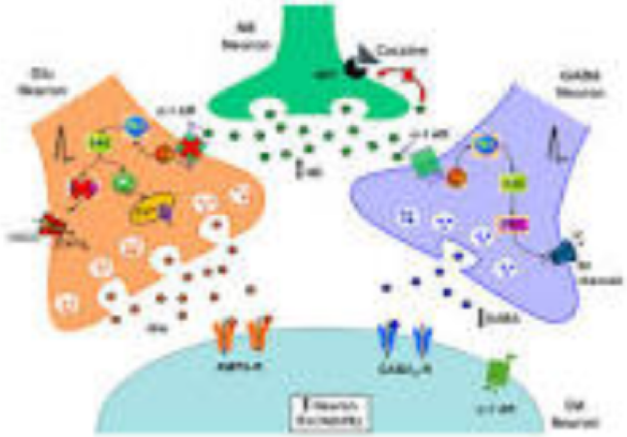
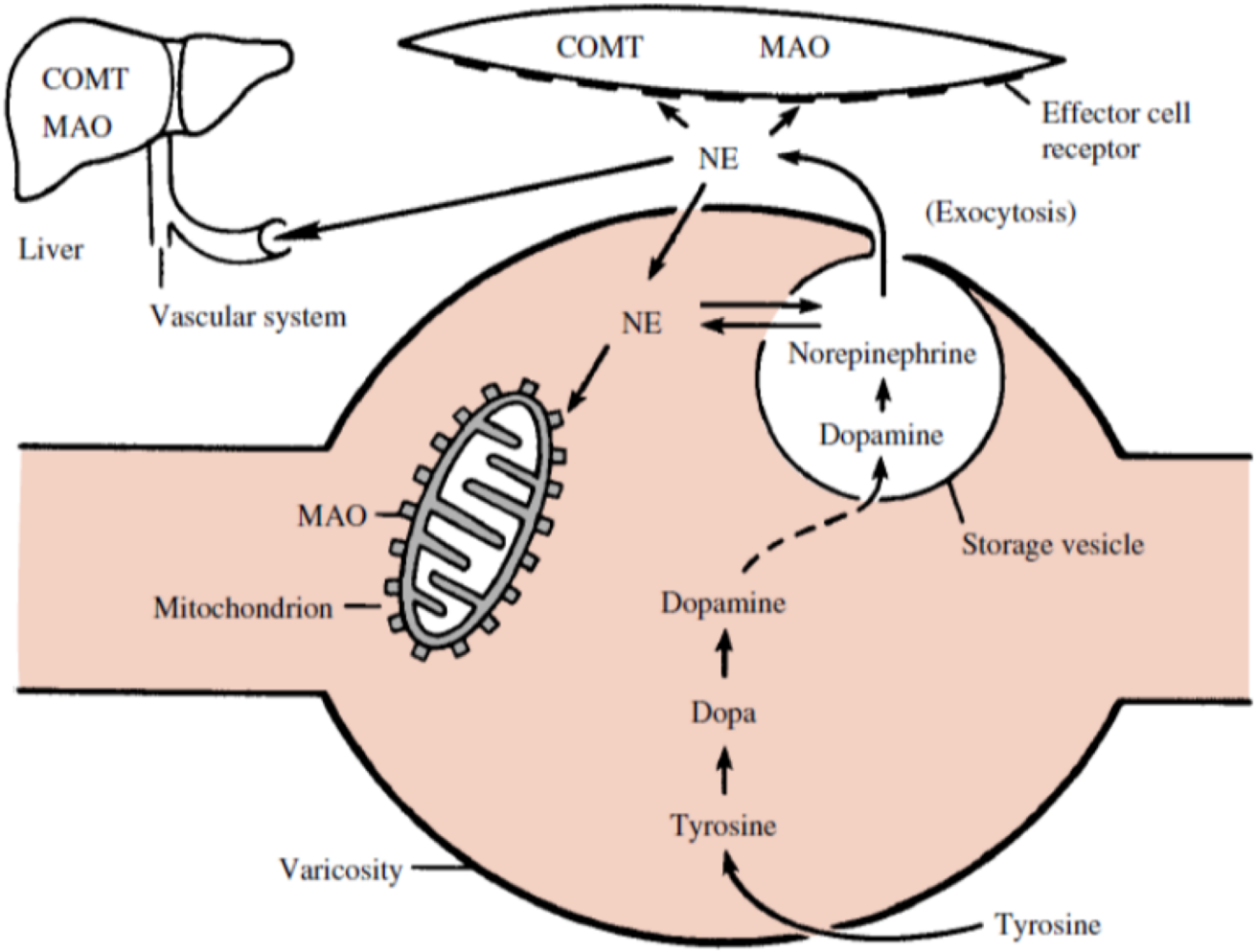
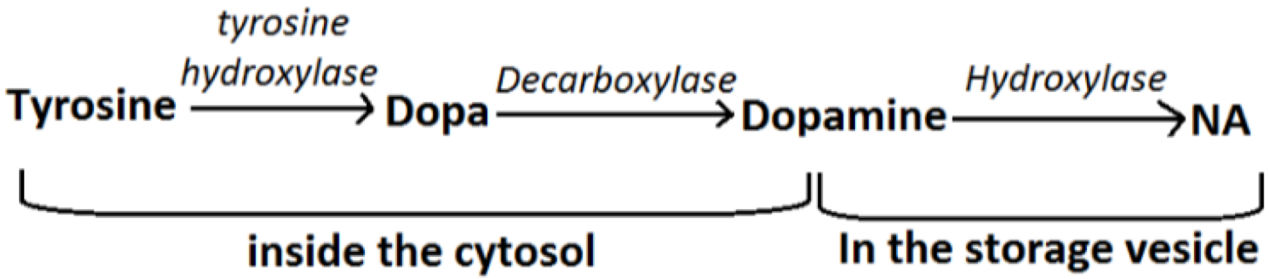


Drugs Affecting Adrenergic Neuron Transmission

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COMT
MAO
COMT
MAO
Effector cell
receptor
NE
(Exocytosis)
Liver
Vascular system
NE
Norepinephrine
Dopamine
MAO
Mitochondrion
Storage vesicle
Dopamine
Dopa
Tyrosine
Varicosity
Tyrosine


tyrosine
hydroxylase
Tyrosine
Dopa
Decarboxylase
Dopamine
Hydroxylase
NA
inside the cytosol
In the storage vesicle

- The released NA (released due to sympathetic stimulation) either

✓ interacts with adrenergic receptors or

✓ enters the circulation and is metabolized in the liver by two enzymes: MAO (monoamine oxidase) & COMT (catechol-O-methyl transferase) or

✓ It might also return to the adrenergic neuron; NA that's been reuptake either enters the storage vesicle or is broken down by MAO.

- Tyrosine hydroxylase is the rate-limiting enzyme. It's inhibited by the end-product of these reactions, the catecholamines: dopamine, NA, and adrenaline.

Drugs affecting adrenergic neuron transmission

- **Metyrosine**: inhibits tyrosine hydroxylase, thereby decreasing L-dopa synthesis and subsequently NA productions. Metyrosine is used to treat pheochromocytoma
- **Reserpine**: inhibits the transfer of dopamine into the vesicle.
- **Amphetamine**: stimulates the release of NA.
- **Selegiline**: inhibits the MAO enzyme.
- **Tolcapone**: inhibits the COMT enzyme.
- **Tricyclic antidepressants** (TCA) and
- **cocaine** inhibit neuronal re-uptake of NA (indirectly acting sympathomimetic agents).

Metyrosine

- Metyrosine works by inhibiting the enzyme **tyrosine hydroxylase,** which is responsible for the conversion of tyrosine to DOPA. By inhibiting the production of DOPA
- metyrosine reduces the levels of catecholamines in the body as adrenalin and noradrenalin.
- This drug is used for the treatment of pheochromocytoma. (a type of hypertension due to the presence of a tumor in the adrenal gland).
- Metyrosine can have many **side effects**, including as **sedation, dry mouth, difficulty urinatung and hypotension.**

α-Methyldopa (Aldomet)

- α-methyldopa is converted by a **decarboxylase enzyme** into **α-methyldopamine**.
- α-methyldopamine is not a good substrate for MAO, so it's not well metabolized by these enzymes.
- This dopamine is then transported to a storage vesicle, where it's converted by a hydroxylase enzyme to **α-methylnorepinephrine**. This occurs centrally.
- α-methylnorepinephrine is an α2 agonist, so when it's released from adrenergic neurons, it binds to α2 receptors, thereby decreasing sympathetic outflow.

- α-methyldopa also competes with L-dopa for the decarboxylase enzyme, thereby causing reduced production of dopamine and, subsequently, NA.
- Its main side effects are sedation, drowsiness, dizziness, and dryness of the mouth.
- Its use is limited due to its side effects and the need for multiple doses a day.
- It's used to **treat hypertension in pregnant women, as it's safe for pregnancy**

Benserazide & Carbidopa:

- **They are decarboxylase enzyme inhibitors.**
- They are used in combination with L-dopa for the treatment of Parkinson's disease.
- Note: Dopamine cannot be used as a drug for Parkinson's because it cannot cross BBB (blood-brain barrier); L-dopa is used as a precursor for dopamine, when L-dopa is administered, most of it is used up and metabolized peripherally. This is where Benserazide and carbidopa come in. They are combined with L-dopa to decrease its side effects and the dose of L-dope that would have otherwise been required.
- Benserazide and carbidopa inhibit the conversion of L-dopa to dopamine peripherally, but not centrally (since they are water-soluble and cannot cross the BBB); this allows most of the L-dopa to be used centrally and minimizes the side effects (caused by peripheral dopamine) of the administered L-dopa.

Reserpine:

- blocks the transfer of dopamine and NA (including reuptake NA) into the vesicles; reserpine is a Catecholamine-depleting Sympatholytic agent (medical sympathectomy).
- This drug used to be used as an anti-hypertensive agent, but has now been largely replaced due to its side effects and the availability of better alternatives.
- Side effects of reserpine include **depression and increased suicidal tendencies**

Guanethidine:

- a drug that prevents exocytosis of NA from the presynaptic membrane (in contrast to amphetamine, which stimulates NA release).
- Guanethidine is used as an anti-hypertensive agent.

Clonidine (α2 agonist):

- **Clinical uses:**
- 1- Withdrawal syndrome of narcotic drugs & tobacco.
- Attention-deficit/hyperactivity disorder (ADHD)
- 2- Prophylactic treatment of migraine.
- 3- Diarrhea in diabetic patients with autonomic neuropathy.
- 4- Menopausal syndrome.
- 5- Neuropathic pain.
- 6- Anxiety and panic disorder.
- 7- Tourette syndrome.
- 8- Clonidine suppression test (use for diagnosis diagnosis ofpheochromocytoma)

Clonidine withdrawal syndrome:

- high-grade fever,
- severe hypertension,
- Headaches
- tachycardia,
- profound sweating
- Anxiety
- Irritability

Note: clonidine needs to be discontinued, the drug should be withdrawn gradually (gradually tapering off).

Guanfacine:

- Alpha 2 receptor agonist
- used for Attention deficit hyperactivity (ADHD) in children.

Brimonidine:

- also an α2 agonist.
- It's the drug of choice for the treatment of glaucoma.
  In the eye, β2 receptor stimulation causes the activation of adenylyl cyclase, which converts ATP to cAMP (second messenger), resulting in the production of aqueous humor. Therefore, beta-blockers such as timolol can decrease intraocular pressure.
  Brimonidine is alpha 2 receptor agonist. Activation of this receptor leads to inhibition of adenyl cyclase, resulting in a decreased production of cAMP subsequently causing a fall in intraocular pressure
-

MAO inhibitors:

- MAO-A is inhibited by **moclobemide** (note: MAO-A metabolizes NA, serotonin, and dopamine)
- MAO-B is inhibited by **rasagiline** (note: MAO-B metabolizes mainly dopamine)
- Rasagiline can be used for Parkinson's disease, as it decreases the breakdown of dopamine. MAO-A inhibitors are used for depression, as they decrease the metabolism of serotonin.

COMT inhibitors:

- **Tolcapone, entacapone,** and **opicapone**

these can also be used for Parkinson's disease.

Tyramine:

- is derived from the amino acid tyrosine.
- It is mainly present in fermented food such as cheese and beverage alcohol.
- Tyramine can enter adrenergic neurons; there it stimulates the release of large amounts of NA.
- Tyramine present in food has no significant effects, because of its destruction (metabolism) by MAO enzymes. However, if it's taken by the body with MAO inhibitors, tyramine is not metabolized in GIT by MAO enzymes, so it enters the adrenergic neurons and stimulates the release of large amounts of NA, causing the hypertensive crisis.

Cocaine and tricyclic antidepressants

- They increase the level of NA and other neurotransmitters in the synapses. Cocaine is a highly addictive stimulant drug.
- Imipramine and amitriptyline.
  They have indirect sympathomimetic effects.
  They inhibit neuronal reuptake of noradrenaline.

Reference

- **Lippincott's Illustrated Reviews Pharmacology 6th edition**